Cardiac Deformation Lab

Advanced Statistical & Quantitative Analytics Report

Study Title:	**Report Test Study 8bf8f4**	**Subject Identifier:**	TEST-REP-01
Study ID:	6edfbf63-20bf-481c-b37c-e16c55c0e04c	**Session Count:**	**2** analyses
Analysis Scope:	**Full Image**	**Detected Outliers:**	0 observations (1.5xIQR)
Report Date:	2026-10-18 09:15 UTC	**Computational Engine:**	NumPy / Farnebäck Dense Flow

1. Descriptive Deformation Statistics (Univariate)

Metric	N	Mean	Median	Std Dev	Min	Max	P25	P75	IQR
Mean Displacement (px)	2	1.5011	1.5011	0.0000	1.5011	1.5011	1.5011	1.5011	0.0000
Max Displacement (px)	2	6.6640	6.6640	0.0000	6.6640	6.6640	6.6640	6.6640	0.0000
Anisotropy Index	2	0.4852	0.4852	0.0000	0.4852	0.4852	0.4852	0.4852	0.0000
Strain Entropy	2	2.8164	2.8164	0.0000	2.8164	2.8164	2.8164	2.8164	0.0000
Strain Gradient	2	0.0329	0.0329	0.0000	0.0329	0.0329	0.0329	0.0329	0.0000
Motion Coherence	2	0.8749	0.8749	0.0000	0.8749	0.8749	0.8749	0.8749	0.0000
Quality Control Score (/100)	2	98.0000	98.0000	0.0000	98.0000	98.0000	98.0000	98.0000	0.0000

2. Longitudinal Change & Standardized Differences

Tracked Metric	First	Latest	Delta (Δ)	Rel Change	Std Diff (d)	Trend
Mean Displacement	1.5011 px	1.5011 px	0.0000 px	+0.0%	0.0000	**Stable**
Max Displacement	6.6640 px	6.6640 px	0.0000 px	+0.0%	0.0000	**Stable**
Anisotropy Index	0.4852	0.4852	0.0000	+0.0%	0.0000	**Stable**
Strain Entropy	2.8164	2.8164	0.0000	+0.0%	0.0000	**Stable**
Strain Gradient	0.0329	0.0329	0.0000	+0.0%	0.0000	**Stable**
Motion Coherence	0.8749	0.8749	0.0000	+0.0%	0.0000	**Stable**
Quality Control Score	98.0000 /100	98.0000 /100	0.0000 /100	+0.0%	0.0000	**Stable**

3. Metric Pearson Correlation Analysis

Insufficient paired observations (<3 sessions) for Pearson metric correlation.

4. Computational Statistical Outliers (1.5×IQR)

No computational outliers identified using the 1.5×IQR boundary rule (or insufficient N < 4 observations).

5. Statistical Warnings & Protocol Consistency

• **Note:** Insufficient paired observations (<3 sessions) for Pearson metric correlation matrix.

6. Research Methodology & Interpretation Disclaimer

Computational Analytics Scope: Descriptive parameters, consecutive change differentials, standardized computational differences (effect sizes), and Pearson correlation matrices reflect algorithmic evaluations of image-space deformation fields under classical continuum kinematics. Statistical outliers reflect mathematical dispersion (1.5×IQR) and do NOT indicate patient health or clinical abnormality.

RESEARCH PROTOTYPE ONLY — NO CLINICAL DIAGNOSTIC ASSERTIONS:

These statistics summarize computational image-derived measurements and should not be interpreted as clinical findings. This software does NOT provide medical diagnoses, prognosis, treatment monitoring, or clinical risk assessments. This document is intended strictly for mathematical and biomechanical research experimentation.